ALI HASSAN

Backend & SaaS Systems Engineer · Automation Specialist

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Executive Profile

Backend and SaaS Systems Engineer specializing in scalable **Node.js** architectures, high-SKU B2B e-commerce catalog pipelines, and production **RAG (Retrieval-Augmented Generation)** automation. Proven track record of replacing brittle third-party middleware with custom sub-50ms Express microservices, architecting zero-hallucination Human-in-the-Loop workflows for complex 30,000+ word technical domains, and engineering production SaaS platforms with custom administrative dashboards (alidigitaltools.com). Adept at building resilient API gateways, bidirectional CRM synchronizations, and modern React back-offices.

Technical Competencies

Backend & SaaS Architecture: Node.js, Express.js, RESTful APIs, Webhook Gateways, Asynchronous Queuing, SSE (Server-Sent Events), Event-Driven Architecture
Database & Storage: MySQL (Sub-5ms Composite Indexing, Soft-Delete Architecture), Schema Sanitization, JSON Pipelines
AI & Automation Systems: Retrieval-Augmented Generation (RAG), Grok xAI, Exemplar-Based Active Learning, Zoho SalesIQ & CRM APIs, IMAP/SMTP Relays
Frontend & Admin Dashboards: React 18, Vite, Tailwind CSS, JavaScript (ES6+), State Management, Administrative Dashboards, Three.js
Operations & Production Delivery: Git, Linux Environments, System Telemetry, Cloud Automation, DNS/SSL Deployment

Engineering Experience

Canadian Timber & Log Enterprise *(International Remote Delivery)* **2025 – Present**
Backend & AI Systems Architect *Production RAG Infrastructure*

- Engineered a centralized multi-channel inbound SaaS engine unifying 5 distinct customer touchpoints (Zoho SalesIQ chat, web forms, CRM scripts, cPanel IMAP mailboxes, and SMS) into an atomic Node.js pipeline.
- Replaced slow, task-metered Zapier/n8n middleware with a custom sub-50ms Express API gateway, eliminating webhook drops and payload schema mismatches during traffic spikes.
- Architected a production RAG pipeline over a 260-chunk, 33,500+ word technical knowledge base, ensuring customer quote responses remain grounded in verified specifications with **0% hallucinations**.
- Designed a **Human-in-the-Loop review portal** with 1-click email approvals, reducing response turnaround **10x** while maintaining 100% human oversight on outbound communications.
- Implemented an exemplar feedback loop where approved human operator adjustments are committed to persistent memory, continuously refining few-shot prompt accuracy.

Ali Digital Tools (alidigitaltools.com) **2024 – Present**
Lead Full-Stack & SaaS Platform Architect *Commercial Platform & Custom Admin Portal*

- Architected and deployed the full-stack commercial web platform and custom administrative back-office dashboard (alidigitaltools.com), managing multi-tier digital subscriptions and catalog discovery.
- Engineered an administrative control portal for customer lifecycle tracking, order state transitions, automated replacement warranty management, and live inventory auditing.
- Built automated transactional fulfillment pipelines integrating webhook relays to dispatch access credentials and order notifications via WhatsApp and Email APIs within sub-minute SLAs.
- Designed an intuitive, high-converting storefront UI using modern React and Tailwind CSS, optimizing checkout UX and real-time inventory queries.

Slashcloud.io **2025 – Present**
Backend & Automation Engineer *Hybrid / Remote*

- Architect cloud-integrated backend services, webhook dispatch processors, and automated API workflows to eliminate manual data entry across enterprise operations.
- Engineer clean asynchronous services with modular error handling, rate limiting, and parameter sanitization to protect edge services against malformed payloads.
- Connect disparate cloud repositories and internal databases via bidirectional sync scripts, slashing operational data latency.

Self-Managed E-Commerce Operations **2024 – Present**
Technical Lead & Operations Founder *PKR 10 Lakh+/Month Scale*

- Bootstrapped and scaled an online commercial platform generating over PKR 10L+/month through data-driven catalog management, conversion funnels, and inventory automation.
- Integrated automated lead capture, order notification webhooks, and customer engagement loops that maximize customer lifetime value (LTV) and order fulfillment velocity.

Featured Architecture Case Study

Enterprise Multi-Channel Human-in-the-Loop AI Orchestration | alistackdev.github.io/case-study.html

- *Architecture:* Node.js, Express, Grok xAI, MySQL Soft-Delete, Zoho SalesIQ & CRM, SSE Real-Time Telemetry.
- Solved multi-turn conversational state tracking and webhook queuing across 5 inbound streams with a custom sub-50ms engine, featuring an interactive architecture diagram and administrative control center.

Education

Government College University Faisalabad (GCUF) — **Faisalabad, Pakistan**
Bachelor of Science in Computer Science (BSCS) — *In Progress*

Relevant Focus: Distributed Database Architectures, Data Structures & Algorithms, Object-Oriented Software Design